Predicting Reaction Products

Predict the products for the following reactions, balance the equation, then classify the type of reaction:

1) ____ Na + ____ $FeBr_3$ □

2) ____ $NaOH$ + ____ H_2SO_4 □

3) ____ C_3H_8 + ____ O_2 □

4) ____ NH_3 + ____ H_2O □

5) ____ $PbSO_4$ + ____ $AgNO_3$ □

6) ____ PBr_3 □

7) ____ HBr + ____ Fe □

8) ____ $KMnO_4$ + ____ $ZnCl_2$ □

9) ____MnO_2 + ____ $Sn(OH)_4$ □

10) ____ O_2 + ____ C_5H_{12} □

11) ____ H_2O_2 □

12) ____ $PtCl_4$ + ____ Cl_2 □

Predicting Reaction Products - Solutions

Balance the equations and predict the products for the following reactions:

1) $\mathbf{3}\ Na + \mathbf{1}\ FeBr_3 \ \square\ \mathbf{3\ NaBr + 1\ Fe}$

2) $\mathbf{2}\ NaOH + \mathbf{1}\ H_2SO_4 \ \square\ \mathbf{1\ Na_2SO_4 + 2\ H_2O}$

3) $\mathbf{1}\ C_3H_8 + \mathbf{5}\ O_2 \ \square\ \mathbf{3\ CO_2 + 4\ H_2O}$

4) $\mathbf{1}\ NH_3 + \mathbf{1}\ H_2O \ \square\ \mathbf{1\ NH_4OH}$

5) $\mathbf{1}\ PbSO_4 + \mathbf{2}\ AgNO_3 \ \square\ \mathbf{1\ Ag_2SO_4 + 1\ Pb(NO_3)_2}$

6) $\mathbf{4}\ PBr_3 \ \square\ \mathbf{1\ P_4 + 6\ Br_2}$

7) $\mathbf{2}\ HBr + \mathbf{1}\ Fe \ \square\ \mathbf{1\ H_2 + 1\ FeBr_2}$ **OR** $\mathbf{6}\ HBr + \mathbf{2}\ Fe \ \square\ \mathbf{3\ H_2 + 2\ FeBr_3}$

8) $\mathbf{2}\ KMnO_4 + \mathbf{1}\ ZnCl_2 \ \square\ \mathbf{2\ KCl + 1\ Zn(MnO_4)_2}$

9) $\mathbf{1}\ MnO_2 + \mathbf{1}\ Sn(OH)_4 \ \square\ \mathbf{1\ Mn(OH)_4 + 1\ SnO_2}$

10) $\mathbf{8}\ O_2 + \mathbf{1}\ C_5H_{12} \ \square\ \mathbf{5\ CO_2 + 6\ H_2O}$

11) $\mathbf{1}\ H_2O_2 \ \square\ \mathbf{1\ H_2 + 1\ O_2}$ **OR** $\mathbf{2}\ H_2O_2 \ \square\ \mathbf{2\ H_2O + 1\ O_2}$

12) $\mathbf{1}\ PtCl_4 + \mathbf{1}\ Cl_2 \ \square\ \mathbf{1\ PtCl_6}$

Predicting Reaction Products (cont.)

For each of the following double replacement reactions, determine what the products of each reaction will be. When you have predicted the products, balance the equation and use a solubility table to determine which of the products (if any) will precipitate (be insoluble). Assume all reactions take place in aqueous solutions (aq).

1) ____ $Ca(OH)_2$ + ____ HF □

2) ____ $Pb(NO_3)_2$ + ____ K_2CrO_4 □

3) ____ $NaC_2H_3O_2$ + ____ H_2SO_4 □

4) ____ $Cu(OH)_2$ + ____ H_3PO_4 □

5) ____ $AgNO_3$ + ____ Na_2CO_3 □

6) ____ $Pb(OH)_2$ + ____ Hg_2S □

Reaction Products Worksheet - Key

For each of the following reactions, determine what the products of each reaction will be. When you have predicted the products, balance the equation and use a table of solubility products to determine which of the products (if any) will precipitate. Remember in order for a chemical change to occur both reactants must be aqueous and one of the products must be leave the solution (as a gas, a precipitate, or form a molecular compound such as H_2O). Check to see that both of the starting reactants are aqueous (soluble in water) or else the reaction will not occur.

1) **1** $Ca(OH)_2$ + **2** HF □ **2 H_2O + CaF_2 (CaF_2 precipitates)**

2) **1** $Pb(NO_3)_2$ + **1** K_2CrO_4 □ **2 KNO_3 + $PbCrO_4$ ($PbCrO_4$ precipitates)**

3) **2** $NaC_2H_3O_2$ + **1** H_2SO_4 □ **Na_2SO_4 + 2 CH_3COOH (no precipitate, thus no rxn)**

4) **3** $Cu(OH)_2$ + **2** H_3PO_4 □ **6 H_2O + $Cu_3(PO_4)_2$**

copper (II) phosphate precipitates

5) **2** $AgNO_3$ + **1** Na_2CO_3 □ **Ag_2CO_3 + 2 $NaNO_3$ (Ag_2CO_3 precipitates)**

6) $Pb(OH)_2$ + Hg_2S □ **no reaction; neither reagent is soluble in water**

